and the possible values of $h = |i-j|$ range from 1 to $n-1$ and we have $n-h$ times each γ_h $\forall h \in [\![1, n-1]\!]$ so:

$$\text{Var}(\bar{Y}_n) = \frac{1}{n^2}\left(n\gamma_0 + 2\sum_{h=1}^{n-1}(n-h)\gamma_h\right)$$

$$\text{Var}(\bar{Y}_n) = \frac{\gamma_0}{n} + \frac{2}{n}\sum_{h=1}^{n-1}\left(1-\frac{h}{n}\right)\gamma_h$$